

Automotive OBD-II GSM + GPS Tracking System

Author: Beer Bhadra Jain

System Overview

Automotive tracking system integrating GSM, GNSS, CAN and embedded control.

Architecture

ESP32, EC200U, MCP2551, MPU6050, DS3231, CP2102.

Power Design

12V input converted to 5V, 4.2V and 3.3V with protection circuits.

GSM Module

EC200U powered with proper decoupling and SIM interface design.

ESP32

Uses EN pull-up resistor and capacitor for stable startup.

CAN Interface

Uses MCP2551 with termination and protection.

Conclusion

Design is functional and near production ready.